

NIRAV SURABHI

West Lafayette, IN · (571) 307-8508 · niravsurabhi@gmail.com · niravsurabhi.com · linkedin.com/in/niravss · github.com/21J3phy

Purdue mechanical engineering student and first-author IEEE researcher who ships production software: a native iOS app on the App Store, real-time voice-AI backends, and instrumented hardware with measured results.

EDUCATION

Purdue University — B.S. Mechanical Engineering (First-Year Engineering) Aug 2026 – May 2030 (expected)

Coursework: ENGR 133, MA 265 Linear Algebra, PHYS 172 Modern Mechanics · **Activities:** Purdue Formula SAE

Honors: Presidential Volunteer Service Award, Silver · Mathematics Scholar, Central Virginia Governor's School · Captain, Science Olympiad · 1st place, Aguado Guitar Competition

EXPERIENCE

I3 Simulations — CAD Intern Jun 2025 – May 2026

- Produced 3D models and haptic-feedback prototypes for medical-training simulators.

The Bike Lane — Bike Mechanic (intern, then part-time) Jun 2021 – Aug 2025

- Assembled bicycles from the frame up and ran the final safety inspection before customer delivery.

Ember Learning — AI Engineer & Software Intern (formerly DeAP Learning Labs) Feb 2021 – 2022

- Wrote the LLM prompts behind AP-tutor chatbots, built automated grading of AP DBQ/FRQ essays against the official rubrics, and indexed lecture transcripts in a vector database for timestamp search; the team secured a **Microsoft and OpenAI grant** and reached a \$1M+ valuation.

RESEARCH & PUBLICATIONS

Passive Vortex Propulsion in Air: A Proof of Concept Demonstration — First Author · IEEE EngiTek 2025 Dec 2025

- Built a wind tunnel and test rig with a soldered, calibrated Arduino load-cell system; measured **3.1 N mean net thrust over 20 trials ($p < 0.001$)**, the first experimental evidence of passive vortex propulsion in air.
- Compared flat and NACA 0024/0012/0009 fins with ANOVA and Dunn's test in Python plus CFD in Fusion 360; won the **AIAA Aerospace Award** at the Regional Science & Engineering Fair (invitation to NASA Goddard) and presented at EngiTek 2025.

PROJECTS

SpareMe — Founder & Lead Engineer · getspareme.com Jul 2026 – Present

- Shipped an iPhone app to the App Store that delegates phone calls to an AI agent: it dials U.S. businesses, works phone trees, waits on hold, and returns a written outcome; users watch a live transcript and can type instructions, listen in, or take over mid-call.
- Built the SwiftUI client and Node/TypeScript backend on Twilio ConversationRelay (real-time speech) and OpenAI (reasoning), with verified caller ID, encrypted PostgreSQL case state, APNs, StoreKit-funded per-minute billing with signed transaction verification, and fail-closed kill switches and rate limits.

FTC Robotics, Absolute Zero #12096 — Captain & Lead Programmer Sep 2022 – Jun 2026

- Led the team for **four seasons**; designed and built every mechanism in Fusion 360 and wrote the autonomous and tele-op code in Java (PID control, sensor fusion, odometry); advanced to the Chesapeake regional championship.

Oh — Creator, open source · github.com/21J3phy/Oh May 2026 – Jun 2026

- Built shared memory for AI coding agents: hook-driven capture of Claude Code and Codex sessions, local secret scrubbing, pgvector retrieval, and an MCP tool that answers why code is the way it is without interrupting its author.

Motor-Based Resistance Trainer — Battery-powered, no-weights strength trainer Mar 2025

- Built three prototypes (eddy-current, chain-driven, geared) of a portable strength trainer; a 188:1 geared motor under Arduino PWM control produced a linear, repeatable force curve ($R^2 = 0.993$) at a \$215 build cost.

SKILLS

Languages: Python, Java, C++, Swift, TypeScript, SQL

Software: SwiftUI, Next.js, Node.js, FastAPI, PostgreSQL, WebRTC, Twilio, StoreKit, APNs, Git, CI/CD

AI / ML: PyTorch, computer vision, LLM agents and tool use, vector search (pgvector), prompt design, evaluation

Hardware: Fusion 360 (CAD, FEA, CFD), 3D printing, Arduino, load cells, soldering, PID / PWM control, sensor fusion